UNIT 1 · 9 hours

Average, RMS, Form and Peak Factor



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

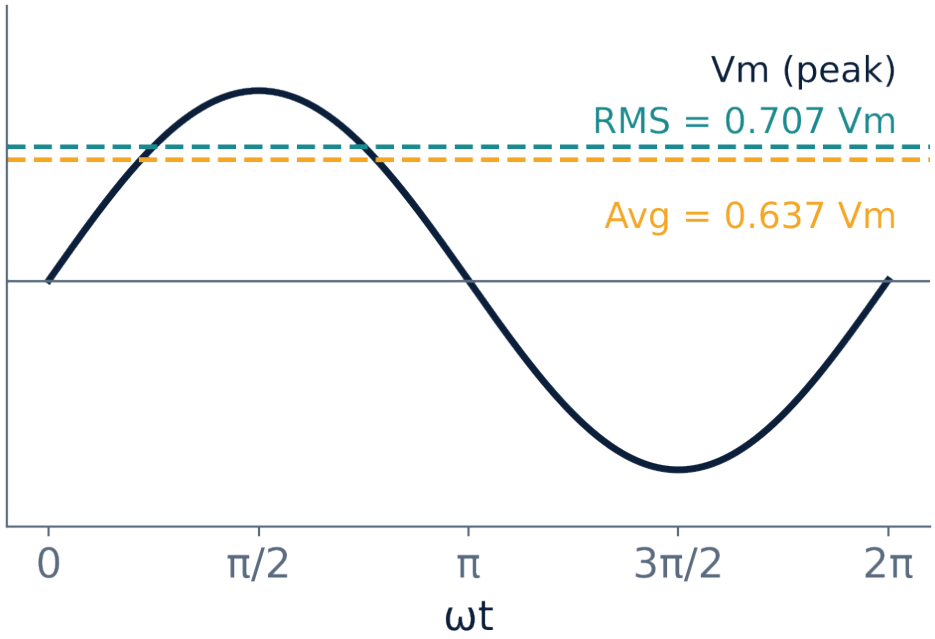
- Derive average and RMS values of a sinusoid
- Explain why RMS is the effective value
- Calculate form factor and peak factor
- Convert between peak, average and RMS

Syllabus context

Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy. Generation of sinusoidal voltage, frequency, average value, RMS value, form factor and peak factor, phasor representation of alternating quantities.

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Average value



- Based on charge transferred (q = I × t).
- Over a full sine cycle the average is zero.
- So average is taken over a half cycle.
- Iav = area / base = $(1/\pi)\int$ Im sin θ dθ
- Iav = 2Im/π = 0.637 Im

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Deriving the RMS value

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Heating

A current i through R produces heat i^2R.

2

Square

Square the instantaneous current: $i^2 = I_m^2 \sin^2\theta$.

3

Mean

Mean of i^2 over half cycle = $(1/\pi)\int I_m^2 \sin^2\theta \, d\theta = I_m^2/2$.

4

Root

Take the square root to get the effective current.

$I_{rms} = I_m / \sqrt{2} = 0.707\, I_m$ and $V_{rms} = V_m / \sqrt{2}$

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Form factor and peak factor

Kf = RMS / Average = 1.11

for a sine wave

Kp = Peak / RMS = 1.414

for a sine wave

- Form factor indicates how peaked or flat the waveform is.
- Peak factor matters for insulation and rating (peak voltage stress).
- Moving-coil meters read average but are marked in RMS using Kf = 1.11.

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Summary of sinusoidal values

Quantity	Formula	Value (Vm or Im = 1)
Average (half cycle)	2 Vm / π	0.637
RMS	Vm / √2	0.707
Form factor	RMS / Avg	1.11
Peak factor	Peak / RMS	1.414

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Example 1: RMS to peak

Problem A sinusoidal current has RMS value 10 A. Find its peak value and average value.

- $I_m = I_{rms} \times \sqrt{2} = 10 \times 1.414 = 14.14$ A
- Average (half-cycle) = 0.637×14.14

Answer: Peak ±14.14 A, average 9.0 A

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Example 2: mains supply

Problem 230 V RMS, 50 Hz mains. Find peak voltage and the voltage 2.5 ms after the zero crossing.

- $V_m = 230 \times \sqrt{2} = 325.27$ V
- $\omega = 2\pi \times 50 = 314$ rad/s; $\theta = 314 \times 0.0025 = 0.785$ rad $= 45°$
- $v = 325.27 \sin 45°$

Answer: $V_m = 325.27$ V; $v = 230$ V

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Key takeaways

- Average (half cycle) = 0.637 × peak
- RMS = 0.707 × peak: effective heating value
- Form factor 1.11 and peak factor 1.414 for a sine
- AC ratings (230 V) are RMS values

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Practice questions

Q1. Derive the RMS value of $i = I_m \sin\theta$.

Q2. A voltage has RMS 110 V. Find its peak and average values.

Q3. A square wave of peak 10 V: what are its RMS and form factor?

Q4. Why is the average of a full sine cycle zero?